Databricks

HUMAN RESOURCES · SOLUTION BRIEF

Hire Right Agent

An AI-powered hiring platform that evaluates every candidate with consistent scoring, transparent recommendations, and natural language analytics — all on a single Databricks lakehouse.

Jackson & Jackson HR Digital · Powered by Databricks

Consistent Evaluation

8-factor weighted scoring eliminates subjective bias and applies uniform criteria to every candidate

Agentic AI

MLflow ResponsesAgent answers complex hiring questions using live data, resumes, and ML predictions

Semantic Resume Search

Vector Search surfaces the most relevant candidate experience for any open role — beyond keyword matching

Natural Language Analytics

Genie Space lets HR leaders query the hiring pipeline in plain English — no SQL required

End-to-End Pipeline

Bronze to gold lakehouse pipeline with data quality monitoring, ML training, and agent deployment in one platform

Governed & Auditable

Unity Catalog ensures every data access, prediction, and recommendation is governed, traceable, and secure

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THE OPPORTUNITY

Hiring is a Data Problem

Most organizations have the data to make great hiring decisions — it's trapped in spreadsheets, disconnected ATS systems, and unstructured resumes. The Hire Right Agent unifies that data on the Databricks lakehouse and adds AI to turn it into action.

Structured Scoring

Eight weighted dimensions applied consistently to every candidate across all open roles

ML-Backed Predictions

Hire/no-hire model trained on historical cohort data — grounded in real outcomes, not intuition

Deep Resume Context

Vector Search retrieves the most semantically relevant resume content for each evaluation

Conversational Analytics

Ask any hiring question in plain English — Genie translates it to SQL and returns live results

Lakeview Dashboards

Interactive hiring analytics with pipeline health, score distributions, and role comparisons

Data Governance

Unity Catalog provides fine-grained access control, PII masking, and full audit lineage

FEATURED APPLICATION

DATABRICKS APP · FIELD ENGINEERING DEMO

Hire Right Agent

A full-stack AI hiring assistant that combines candidate scoring, agentic recommendations, resume search, and Genie analytics — deployed as a Databricks App with a FastAPI backend and a responsive web frontend.

MLflow ResponsesAgent · Vector Search · Genie Space · Model Serving · Databricks Apps · Unity Catalog

AI-POWERED HIRING ASSISTANT

Hire Right Agent

Evaluate every candidate with consistent AI scoring, explainable recommendations, and natural language analytics — all in one unified platform.

THE CHALLENGE

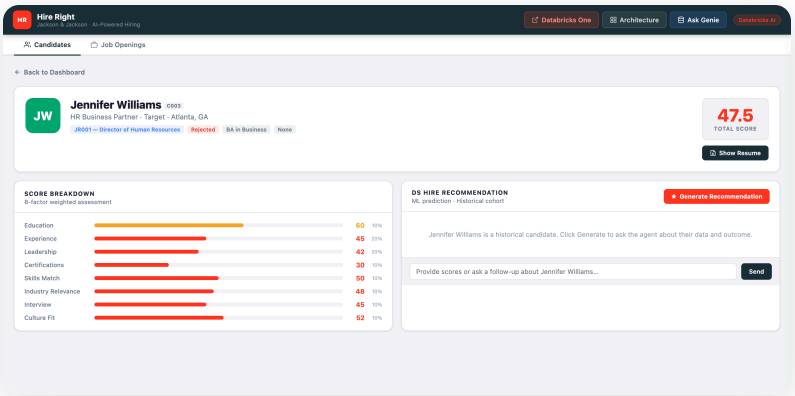
- Candidate reviews are slow, inconsistent, and rely on subjective judgment without structured criteria
- Historical hiring outcomes are siloed — teams can't learn from past decisions to improve future ones
- No single workflow connects data analytics, AI recommendations, and hiring manager communication
- Resume screening is keyword-based and misses qualified candidates with different backgrounds

KEY CAPABILITIES

- → 8-factor weighted scoring with full transparency and per-factor breakdowns
- → AI agent answers natural language questions about any candidate or role
- → Semantic resume retrieval surfaces the most relevant experience — not just keywords
- → ML model predicts hire/no-hire based on historical cohort patterns
- → Genie Analytics for live SQL-backed pipeline and score queries
- → One-click email to hiring managers with AI-composed summaries

DATABRICKS PLATFORM

Databricks Apps · MLflow ResponsesAgent · Vector Search · Genie Space · Model Serving · Unity Catalog · Lakeview Dashboard · MLflow Tracking · Delta Lake · Asset Bundles



AI-POWERED INSIGHT

The Hire Right Agent uses an **MLflow ResponsesAgent** deployed on Databricks Model Serving. It calls four specialized tools — Genie for live analytics, Vector Search for resume context, an ML endpoint for hire/no-hire predictions, and Mailgun for email — delivering fully grounded, explainable recommendations that HR leaders can act on immediately.

20 Candidates
Evaluated across 4 open roles with consistent, auditable AI scoring in a single platform

8-Factor AI
Weighted scoring across education, experience, leadership, skills match, culture fit, and more

Full Stack
Raw data to agentic hiring recommendation — built entirely on Databricks, no external AI stack required

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TECHNICAL REFERENCE

Platform Architecture

Hire Right Agent is built entirely on the Databricks Data Intelligence Platform — from raw data ingestion to agentic AI. Unity Catalog governs every layer.

🔒 Unity Catalog · bx4.hrd_2030

Unified governance, fine-grained access control, PII masking, and full audit lineage across all data and AI assets

DATA SOURCES

Candidate Data

- 20 candidate profiles (CSV)
- 4 open job requirements (CSV)
- PDF resumes (Databricks Volume)

Historical Outcomes

- 10 historical hires & rejects
- 8-factor score labels
- Training data for ML model

External Integrations

- Mailgun (email delivery)
- Databricks Model Serving

DATABRICKS PLATFORM

Lakehouse Pipeline (Delta Lake)

- Bronze → Silver → Gold medallion architecture
- Data quality monitoring & PII masking
- Business semantics & candidate scoring
- Orchestrated via Databricks Jobs

AI & ML Layer

- MLflow: experiment tracking & model registry
- Hire/no-hire classifier (Model Serving)
- Resume embeddings (Vector Search index)
- ResponsesAgent with 4 tool calls
- Agent endpoint (Model Serving)

Analytics & Governance

- Genie Space: natural language SQL analytics
- Lakeview AI/BI Dashboard
- Unity Catalog: catalog · schema · volumes
- Deployed with Asset Bundles (IaC)

APPLICATIONS

Hire Right App

- Databricks Apps deployment
- FastAPI backend (Python)
- Responsive HTML/JS frontend
- Candidate grid & detail views
- Resume PDF viewer

AI Agent

- MLflow ResponsesAgent
- query_genie → HR analytics
- search_resumes → Vector Search
- predict_score → ML endpoint
- send_email → Mailgun

Analytics

- Genie Space (conversational)
- Lakeview Dashboard (visual)
- MLflow Traces (agent observability)

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